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# ContrastiveVAE: Robust Representation Learning for Cross-Sectional Asset Pricing

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## Abstract

1        Extracting structural macroeconomic signals from financial time-series data re-  
2        quires navigating a notoriously low signal-to-noise ratio. While Variational Autoen-  
3        coders (VAEs) offer powerful representation capabilities, standard implementations  
4        frequently overfit to idiosyncratic volatility in cross-sectional asset pricing. In this  
5        paper, we propose a Contrastive VAE architecture. We modify the FactorVAE [1]  
6        by upgrading its temporal feature extractor with a Pre-Layer Normalized Trans-  
7        former and Gated Linear Units (GLUs) to stabilize gradient flow and dynamically  
8        filter input noise. Our primary contribution, however, is the integration of a Cross-  
9        Sectional Contrastive Learning (CSCL) framework to optimize the latent space. By  
10       applying a symmetric InfoNCE loss to the deterministic prior means of randomly  
11       subsampled market views, we enforce permutation invariance across the stock  
12       universe. This self-supervised alignment circumvents the probabilistic conflict of  
13       standard VAE-contrastive integrations and establishes an information bottleneck,  
14       forcing the network to map varying market subsets into unified, regime-aware  
15       representations. Evaluated on a high-dimensional universe of technical indicators,  
16       this architecture successfully isolates structural market factors, yielding highly  
17       separable latent spaces that significantly improve Rank Information Coefficient  
18       (IC) and algorithmic trading performance.

## 19    1 Introduction

20    At its core, cross-sectional asset pricing is fundamentally a representation learning problem. Tra-  
21    ditionally, this domain has been governed by linear factor models derived from Arbitrage Pricing  
22    Theory (APT). In the classical paradigm, a stock’s expected return is modeled as a linear combination  
23    of its exposures (loadings) to a small, predefined set of systemic risk factors—such as value, size,  
24    or momentum. By reducing thousands of equities into a handful of macroeconomic drivers, factor  
25    models provide a critical framework for risk management and alpha generation.

26    However, these traditional linear models struggle to capture the complex, time-varying, and highly  
27    non-linear interactions inherent in modern financial markets. Deep learning reframes this paradigm:  
28    rather than relying on manually constructed, static factors, advanced neural architectures can au-  
29    tonomously discover a low-dimensional latent space. The objective is to map a high-dimensional,  
30    highly volatile universe of stock characteristics into this latent space, accurately isolating structural  
31    macroeconomic drivers while aggressively filtering out idiosyncratic noise.

32    Achieving this separation is critical for algorithmic trading, yet it remains exceptionally difficult  
33    because financial time-series data is characterized by an inherently low signal-to-noise ratio. To  
34    fulfill the critical need for noise-resistant factor extraction, we propose the **ContrastiveVAE**. This  
35    specialized, self-supervised encoder-decoder architecture is explicitly engineered to navigate the  
36    extreme noise environment of quantitative finance. By introducing deliberate architectural bottlenecks

and latent space constraints, we provide a mathematically rigorous framework for isolating slow-moving structural factors from rapid, localized market volatility.

Specifically, our contributions to the domain of deep learning for asset pricing are threefold:

- **Dynamic Noise Filtration and Stabilized Temporal Attention:** We redesign the temporal feature extractor in the FactorVAE [1] by preceding the attention mechanism with Gated Linear Units (GLUs), allowing the network to dynamically mute irrelevant technical indicators. Furthermore, we replace standard post-normalization with Pre-Layer Normalization (`norm_first=True`) and introduce a learnable [CLS] token to guarantee smooth gradient flow and prevent signal dilution.
- **Cross-Sectional Contrastive Learning (CSCL):** We introduce a CSCL framework that subsamples distinct, randomized cross-sections of the market for the same trading day. This forces the network to reconstruct identical market states from heterogeneous data, strictly enforcing permutation invariance across the stock universe.
- **Deterministic Symmetric InfoNCE:** We resolve the architectural conflict between variational inference and contrastive optimization by applying a symmetric contrastive penalty strictly to the deterministic prior means ( $\mu_{prior}$ ). This establishes an information bottleneck that groups structurally similar market regimes without corrupting the generative capabilities of the VAE.

## 2 Background and Related Work

In quantitative finance, the transition towards advanced factor representations was catalyzed by foundational frameworks such as Instrumental Principal Component Analysis (IPCA) [3], which successfully modeled dynamic, time-varying factor loadings using observable firm characteristics. Building upon this, subsequent literature introduced deep autoencoder architectures [2] to capture complex non-linearities in both asset exposures and macroeconomic latent factors.

Recently, these approaches have evolved into generative frameworks, with models like the FactorVAE [1] attempting to unify deep probabilistic representation learning with traditional asset pricing mechanics. While the foundational FactorVAE represented a significant leap forward, it exhibited notable architectural limitations. First, its reliance on Gated Recurrent Units (GRUs) for temporal feature extraction inherently restricted its capacity to model complex, long-range sequential dependencies, presenting a strong potential for improvement by upgrading to a modern Transformer-based attention mechanism. Second, the original architecture was brittle when faced with cross-sectional variability; it lacked the ability to reliably differentiate or align market samples containing lower numbers of available stocks that nevertheless belonged to the exact same underlying macroeconomic regime; which we tackle through a contrastive learning framework.

## 3 Methods and Model Architecture

The ContrastiveVAE is a self-supervised, probabilistic encoder-decoder network. The architecture is designed to map a multi-variate time-series of cross-sectional stock characteristics into a low-dimensional, highly separable latent factor space. In this section, we detail the mathematical formulation of the model, explicitly distinguishing between the foundational mechanics derived from the original FactorVAE literature and our novel architectural interventions.

### 3.1 Problem Formulation

Let  $N$  denote the number of available stocks in the cross-section at time  $t$ . For each stock  $i \in \{1, \dots, N\}$ , we observe a time-series of technical characteristics over a lookback window  $T$ , yielding an input tensor  $\mathbf{X}_t \in \mathbb{R}^{N \times T \times C}$ , where  $C$  is the number of features. The objective is to predict the cross-sectional asset returns  $\mathbf{y}_{t+1} \in \mathbb{R}^N$  by extracting  $K$  latent macroeconomic factors, where  $K \ll C$ .

### 83 3.2 Temporal Feature Extraction (Novel Contribution)

84 The original FactorVAE utilizes standard Recurrent Neural Networks (GRUs) combined with simple  
 85 linear layers to process the temporal dimension of stock characteristics. However, in low signal-to-  
 86 noise environments, GRUs struggle with long-range dependencies, and standard linear projections  
 87 are highly susceptible to collinearity among technical indicators. We completely replace this module  
 88 with a novel, dynamically filtered attention mechanism.

89 **1. Dynamic Noise Filtration via GLU:** Before temporal processing, the raw input  $\mathbf{X}$  is passed  
 90 through a Gated Linear Unit (GLU). The GLU acts as a learnable, non-linear feature selector:

$$Gate(\mathbf{X}) = (\mathbf{X}\mathbf{W}_1 + \mathbf{b}_1) \odot \sigma(\mathbf{X}\mathbf{W}_2 + \mathbf{b}_2) \quad (1)$$

91 where  $\odot$  is the element-wise Hadamard product and  $\sigma$  is the sigmoid activation. To prevent "dead  
 92 gates" early in training, the bias vector  $\mathbf{b}_2$  is initialized to 1.0. This allows the network to dynamically  
 93 attenuate noisy or redundant technical features on a per-stock basis before they contaminate the  
 94 attention mechanism.

95 **2. Pre-Layer Normalized Transformer with [CLS] Aggregation:** The filtered representations are  
 96 projected into a hidden dimension  $H$  and prepended with a learnable [CLS] token. We deploy a multi-  
 97 layer Transformer Encoder to model the temporal dynamics. Critically, we eschew the default PyTorch  
 98 Post-Layer Normalization architecture in favor of **Pre-Layer Normalization** (`norm_first=True`).  
 99 In Pre-LN, the normalization is applied inside the residual blocks:

$$\mathbf{h}' = \mathbf{h} + \text{MultiHeadAttention}(\text{LayerNorm}(\mathbf{h})) \quad (2)$$

$$\mathbf{h}'' = \mathbf{h}' + \text{FeedForward}(\text{LayerNorm}(\mathbf{h}')) \quad (3)$$

100 This architectural decision guarantees smooth gradient flow to the lower layers from the very first  
 101 epoch, preventing the gradient collapse frequently observed when applying vanilla Transformers to  
 102 volatile financial data. The final representation for stock  $i$ , denoted as  $e_i$ , is extracted exclusively  
 103 from the [CLS] token's output state, effectively summarizing the temporal trajectory without the  
 104 signal dilution caused by standard mean-pooling.

### 105 3.3 Factor Encoder / The Oracle ( $\phi_{enc}$ )

106 The Factor Encoder compresses the actual future returns  $y \in \mathbb{R}^N$  into the optimal posterior factors.  
 107 To prevent overfitting and manage the high-dimensional cross-section, we utilize a dynamic portfolio  
 108 sorting mechanism.

1. **Dynamic Portfolio Layer:** We map the  $N$  stocks into  $M$  distinct "portfolios" ( $M \ll N$ ).  
 A softmax layer generates the portfolio weights  $W_{port} \in \mathbb{R}^{N \times M}$  based on the stock features  
 $e$ :

$$W_{port} = \text{Softmax}(eW_P + b_P)$$

The future returns are then aggregated into portfolio returns  $y_{port} \in \mathbb{R}^M$ :

$$y_{port} = W_{port}^\top y$$

2. **Gaussian Parameterization:** A Multi-Layer Perceptron (MLP) processes  $y_{port}$  to output  
 the parameters of the posterior distribution. A Softplus activation ensures the standard  
 deviation remains strictly positive:

$$\begin{aligned} \mu_{post} &= \text{Linear}_\mu(\text{MLP}(y_{port})) \\ \sigma_{post} &= \text{Softplus}(\text{Linear}_\sigma(\text{MLP}(y_{port}))) \end{aligned}$$

### 109 3.4 Factor Predictor / The Prior ( $\phi_{pred}$ )

110 The Predictor extracts the global market factors using exclusively historical data  $e$ . It utilizes a  
 111 cross-sectional attention mechanism to dynamically weight the importance of different stocks.

1. **Multi-Head Global Attention:** We initialize a learnable query matrix  $Q_{global} \in \mathbb{R}^{K \times d_{model}}$ .  
 The keys  $K$  and values  $V$  are linear projections of the stock features  $e$ . The attention  
 mechanism calculates the influence of each stock on the  $K$  global factors:

$$\text{Attention}(Q_{global}, K, V) = \text{Softmax}\left(\frac{Q_{global}K^\top}{\sqrt{d_{model}}}\right)V$$

We use  $H$  attention heads, concatenating their outputs to capture diverse systemic risk premiums. Let the aggregated output be  $h_{global} \in \mathbb{R}^K$ .

2. **Prior Distribution Network:** An MLP maps the global representation to the prior factor distribution:

$$\begin{aligned}\mu_{prior} &= \text{Linear}_{\mu}(\text{MLP}(h_{global})) \\ \sigma_{prior} &= \text{Softplus}(\text{Linear}_{\sigma}(\text{MLP}(h_{global})))\end{aligned}$$

### 3.5 Factor Decoder ( $\phi_{dec}$ ) and Joint VAE Objective

During the forward pass, latent factors  $z \in \mathbb{R}^K$  are sampled using the reparameterization trick:  $z = \mu + \sigma \odot \epsilon$ , where  $\epsilon \sim \mathcal{N}(0, I)$ . The Decoder maps these systemic factors and the individual stock features  $e$  to the final return distribution, outputting the predicted cross-sectional returns  $P_{\phi_{dec}}(\hat{y} \mid x, z_{post})$ .

The base generative network is optimized by maximizing the Evidence Lower Bound (ELBO), which decomposes into the Negative Log-Likelihood (NLL) of the reconstructed returns and the Kullback-Leibler (KL) Divergence between the posterior and prior distributions to drive the learning in our predictions:

$$\mathcal{L}_{VAE} = -\frac{1}{N} \sum_{i=1}^N \log P_{\phi_{dec}}(\widehat{y}_{rec}^{(i)} = y^{(i)} \mid x, z_{post}) + \gamma \text{KL}(P_{\phi_{enc}}(z \mid x, y), P_{\phi_{pred}}(z \mid x)) \quad (4)$$

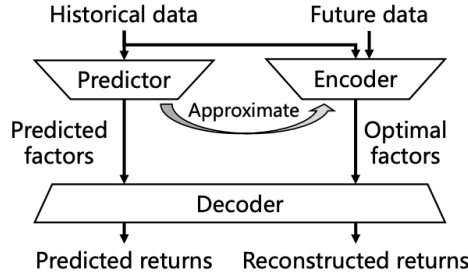


Figure 1: Architecture of the ContrastiveVAE

### 3.6 Cross-Sectional Contrastive Learning (Novel Contribution)

While the VAE effectively models the probabilistic nature of asset returns, it lacks the inductive bias to recognize that a macroeconomic regime remains consistent regardless of the specific subset of stocks observed. To strictly enforce **permutation and subset invariance**, we introduce a novel Cross-Sectional Contrastive Learning (CSCL) framework.

For a given trading day  $t$  containing  $N$  total available stocks, we randomly sample two disjoint subsets, generating View 1 ( $\mathbf{X}^{(1)}$ ) and View 2 ( $\mathbf{X}^{(2)}$ ). Both views are passed through the Feature Extractor and Factor Predictor.

**The Deterministic Bottleneck:** A major failure mode in integrating contrastive learning with VAEs occurs when the InfoNCE loss is applied to the stochastically sampled latent vector  $\mathbf{z}$ . The reparameterization noise ( $\epsilon$ ) violently scrambles the angular relationships required by Cosine Similarity. To solve this, our CSCL loss bypasses the sampling layer entirely and is anchored exclusively to the deterministic mean vector of the prior:  $\mu_{prior}^{(1)}$  and  $\mu_{prior}^{(2)}$ .

**Symmetric InfoNCE Loss:** We accumulate a batch of  $B$  trading days, generating  $2B$  total views. For any given anchor view, the positive sample is the alternate view from the *same* day, while the negative samples are the views from all *other* days in the batch. We optimize the latent geometry using a Symmetric InfoNCE loss:

$$\ell(u, v) = -\log \frac{\exp(\text{sim}(u, v)/\tau)}{\sum_{j=1}^B \exp(\text{sim}(u, k_j)/\tau)} \quad (5)$$

where  $\text{sim}(\cdot, \cdot)$  is the cosine similarity and  $\tau$  is the temperature parameter. To prevent asymmetrical weight updates in the shared Transformer, the final contrastive objective computes the loss bidirectionally:

$$\mathcal{L}_{CSCL} = \frac{1}{2B} \sum_{i=1}^B [\ell(\mu_i^{(1)}, \mu_i^{(2)}) + \ell(\mu_i^{(2)}, \mu_i^{(1)})] \quad (6)$$

### 3.7 Total Joint Optimization

The entire architecture is trained end-to-end via a multi-task learning objective. The contrastive loss acts as a powerful self-supervised regularizer, establishing an information bottleneck that forces the  $\mu_{prior}$  space to cluster by macroeconomic regime before the VAE samples from it:

$$\mathcal{L}_{Total} = \mathcal{L}_{VAE} + \lambda \mathcal{L}_{CSCL} \quad (7)$$

where  $\lambda$  controls the weight of the contrastive penalty.

## 4 Empirical Results

The model was validated using technical features engineered from the Alpha158 dataset on US market data from Yahoo Finance. Training spanned from 2010 to 2016, with out-of-sample testing conducted on the 2018–2019 period.

**Predictive Capability:** The model achieved a Rank Information Coefficient (Rank IC) of 0.038, demonstrating a statistically significant ability to rank cross-sectional returns relative to market baselines.

**Portfolio Generalization (TDrisk):** We deployed the model’s probabilistic outputs using the TDrisk portfolio strategy. Instead of ranking stocks by expected return ( $\mu$ ), stocks were ranked by a risk-adjusted score:  $\text{Score} = \mu_{pred} - \eta \sigma_{pred}$ . We utilized a daily-rebalanced Top-50 equal-weight portfolio, simulating real-world transaction constraints (max turnover of 5 stocks per day).

Table 1: Out-of-Sample Portfolio Backtest Performance

| Strategy                 | Annualized Return | Sharpe Ratio | Max Drawdown |
|--------------------------|-------------------|--------------|--------------|
| FactorVAE (TDrisk Score) | 17.34%            | 1.73         | 6.53%        |

The TDrisk configuration achieved a highly asymmetric risk-return profile. The low Maximum Drawdown validates the theoretical premise: penalizing stocks with high predicted variance successfully immunizes the portfolio against catastrophic noise events.

**Contributions:** As the sole contributor to this project, I focused on the end-to-end implementation of the probabilistic architecture, developing the transformer and contrastive framework, the integration of the mathematical loss functions, and the simulation of the backtesting environment.

## 5 Conclusions and Future Work

This work demonstrates that modeling latent financial factors as conditional probability distributions provides a decisive statistical edge over deterministic deep learning frameworks. By integrating dynamic factor modeling with a Variational Autoencoder, we can quantify epistemic uncertainty and build highly robust, risk-adjusted portfolios that survive low signal-to-noise environments.

**Future Work:** The current framework relies on the assumption that latent factors follow a Gaussian distribution. Financial market returns, however, are known to exhibit skewness and fat tails. Future research will explore replacing the standard Gaussian prior with Normalizing Flows, allowing the VAE to map simple distributions into highly complex, non-linear manifolds that better represent true macroeconomic phenomena.

175 **6 Code and Recordings**

176 Github Repository

177 YouTube Recording

## References

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